

Automated Invoice Processing and Anomaly Detection

Technical Project Report

1. Project Overview

This project implements an end-to-end Machine Learning system for automated invoice processing and anomaly detection. Invoice images are processed using OpenCV, converted into text using Tesseract OCR, parsed into structured invoice and line-item information, validated using financial consistency rules, and analyzed for unusual records using statistical and unsupervised Machine Learning techniques.

2. Problem Statement

Manual invoice processing requires significant time and may introduce errors during data entry, calculation, verification, and duplicate identification. The proposed system automates invoice information extraction and performs validation and anomaly screening to assist invoice-processing workflows.

3. System Architecture

Stage	Technology / Module	Purpose
Image preprocessing	OpenCV / preprocessing.py	Grayscale conversion, noise reduction and binarization
OCR	Tesseract / ocr_engine.py	Convert invoice image into machine-readable text
Information extraction	Regex / parser.py	Extract invoice, item and financial fields
Validation	validation.py	Check financial consistency rules
Anomaly detection	Isolation Forest / anomaly_detector.py	Identify unusual multivariate invoice records
Pipeline	main.py	Connect processing components

4. Modular Codebase

The project follows a modular Python architecture. **preprocessing.py** performs image cleanup, **ocr_engine.py** performs OCR, **parser.py** extracts structured invoice fields, **anomaly_detector.py** contains the Isolation Forest implementation, **validation.py** performs financial consistency checks, and **main.py** connects the processing workflow.

5. Information Extraction

OCR converts the visual content of an invoice into machine-readable text. Regular Expressions (Regex) are then used to locate structured patterns within the OCR output. The parser extracts Invoice Number, Invoice Date, Vendor Name, Total Amount, Description, Quantity, Unit Price, Net Worth, VAT and Gross Worth.

Different invoice layouts can arrange financial information differently. The parser therefore includes handling for several financial layouts encountered in the dataset. The JSON information provided with the dataset was used only as reference ground truth for evaluation and was not used to fill production output fields.

6. Verified Production Results

Metric	Result
Invoice images processed	499
OCR processing coverage	100%
Invoices with extracted items	486 / 499 (97.39%)
Extracted item records	1,805
Quantity extraction	1,788 / 1,805 (99.06%)
Unit Price extraction	1,574 / 1,805 (87.20%)
Net Worth extraction	1,574 / 1,805 (87.20%)

VAT extraction	1,574 / 1,805 (87.20%)
Gross Worth extraction	1,699 / 1,805 (94.13%)
Records processed by production pipeline	1,805
Valid financial records	474
Invalid or incomplete financial records	1,331

7. Rule-Based Validation

Rule-based validation checks whether extracted financial values are internally consistent. For each line item, Quantity multiplied by Unit Price is compared with Net Worth. Net Worth and VAT are then used to calculate the expected Gross Worth. A tolerance of 0.01 is used to account for normal decimal rounding.

Additional checks include invoice-date logic and duplicate detection. These rules help distinguish records that are mathematically consistent from records that require further investigation.

The current production output contains 474 valid financial records and 1,331 invalid or incomplete records. An invalid validation result does not automatically indicate fraud because missing or incorrectly extracted OCR fields can cause financial consistency checks to fail.

8. Anomaly Detection Using Isolation Forest

Isolation Forest is an unsupervised Machine Learning algorithm designed to identify unusual observations. It works by repeatedly partitioning observations; unusual observations generally require fewer partitions to isolate.

Isolation Forest was selected because the project does not have reliable pre-labelled fraud data. It can identify unusual multivariate numerical behavior and is suitable for screening potentially anomalous invoice records.

The current production pipeline generated anomaly predictions for 1,805 item-level records using Quantity, Unit Price, Net Worth and Gross Worth. The output identified 1,543 records as normal and 262 records as potential anomalies, corresponding to an anomaly rate of 14.52%. These records represent potential anomalies and should not automatically be interpreted as confirmed fraud.

9. Statistical and Duplicate Checks

Check	Result
Vendor Z-score	499 checked; 6 had sufficient repeated-vendor history; 0 reached $ Z \geq 3$
Date logic	20 missing/invalid; 468 older than 90 days; 0 future dates
Exact duplicate invoice numbers	0
Exact duplicate amounts	10
Fuzzy duplicate invoice numbers	0
Fuzzy duplicate amounts	13
Total potential duplicate records	21

10. Investigation Cases

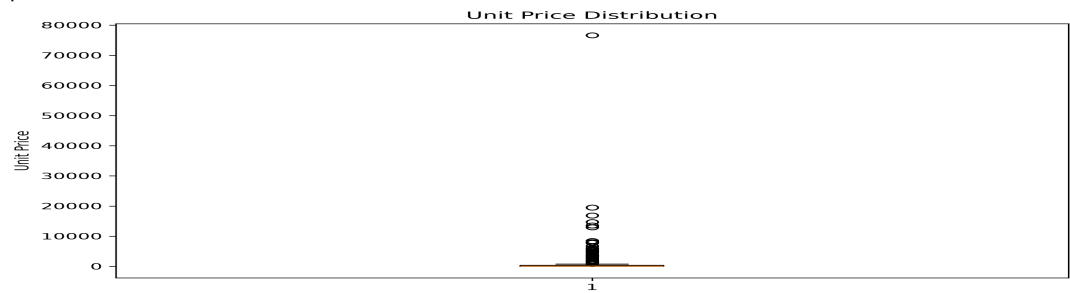
Four records require further investigation because their extracted financial values show stronger inconsistencies or unusual patterns.

Invoice	Item	Quantity	Unit Price	Net Worth	Gross Worth
66053252	7	0	45150.30	4515.03	1315.51
60665075	5	25	174.25	4348.50	4783.35
97781335	2	2	3600.00	7200.00	920.00
52230192	1	30	17.50	517.50	569.25

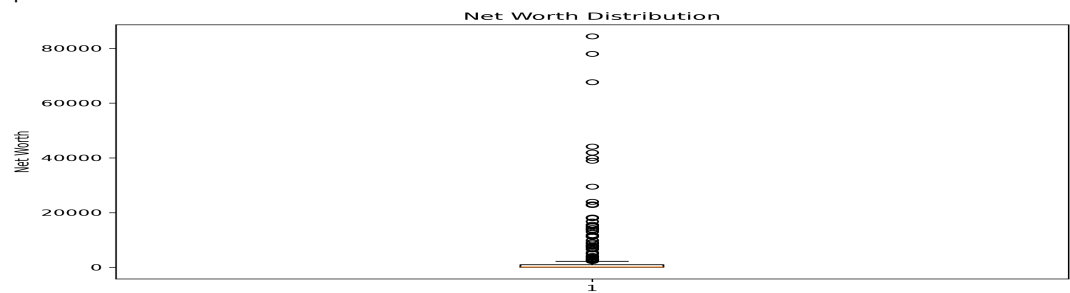
Two negative VAT values remain in the production data. They were not silently corrected because the available OCR evidence is insufficient to determine the intended values. They should remain flagged for manual review.

11. Visual Analysis

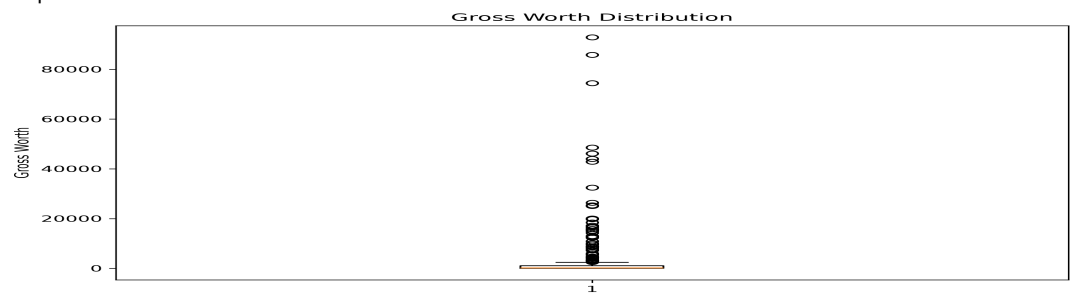
Unit Price Boxplot



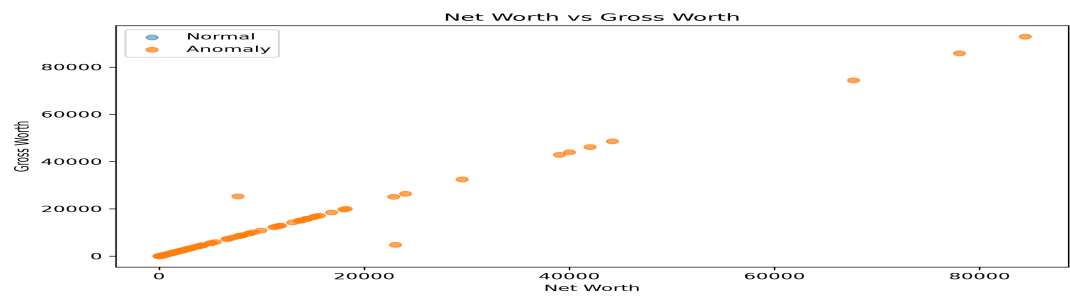
Net Worth Boxplot



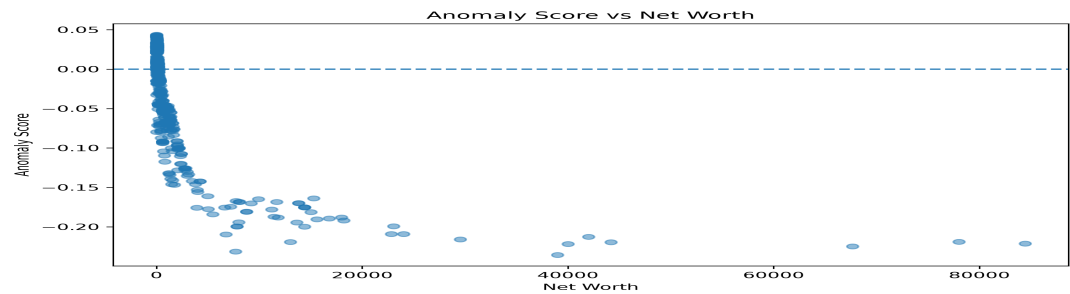
Gross Worth Boxplot



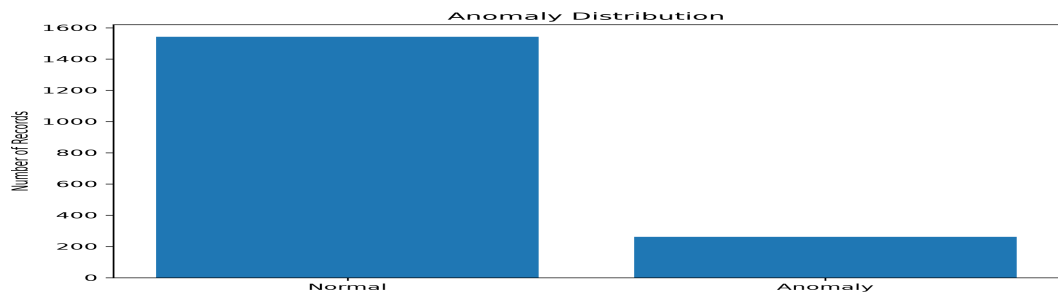
Net Worth Vs Gross Worth Anomalies



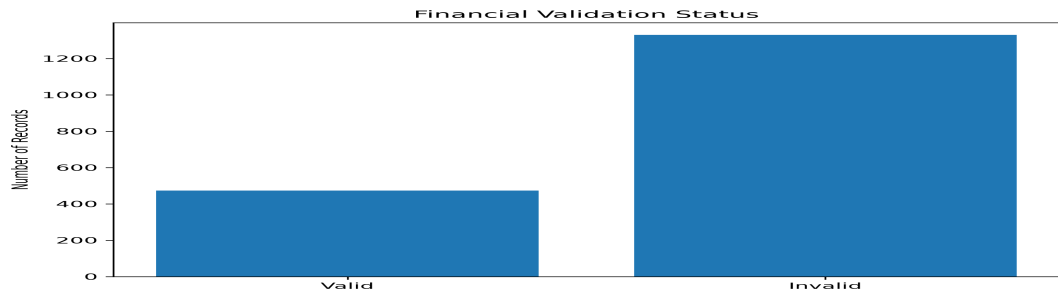
Anomaly Score Vs Net Worth



Anomaly Distribution



Anomaly Validation Status



12. Limitations

The production validation results include invalid or incomplete records caused by missing or inaccurate OCR extraction. Such records require review and should not be treated as confirmed fraudulent invoices.

OCR accuracy depends on invoice image quality and document layout. Different invoice templates can produce different extraction results. Some financial fields may remain unavailable because of OCR errors or layout differences.

Vendor-level Z-score analysis is limited when vendors have insufficient historical records. Isolation Forest requires complete numerical features for model analysis. Fuzzy duplicate detection identifies potential similarity and therefore requires human confirmation.

Anomaly detection identifies unusual behavior and does not by itself prove fraud.

13. Conclusion

The project demonstrates an automated invoice-processing workflow combining Computer Vision, OCR, Regex-based information extraction, rule-based validation, statistical analysis, duplicate detection and unsupervised Machine Learning.

The verified production workflow processed 499 invoice images and produced 1,805 item-level records. The current production anomaly detection output identified 1,543 normal records and 262 potential anomalies, corresponding to an anomaly rate of 14.52%.

The modular Python architecture separates image preprocessing, OCR, parsing, validation, anomaly detection and pipeline execution, satisfying the required modular codebase structure.

14. Final Deliverables

The repository contains preprocessing.py for image preprocessing, ocr_engine.py for OCR, parser.py for information extraction, anomaly_detector.py for the Machine Learning implementation, validation.py for financial validation, and main.py for pipeline integration.

The verified production CSV is stored at **reports/final_invoice_output.csv**. Project visualizations are stored in **reports/visualizations/**.